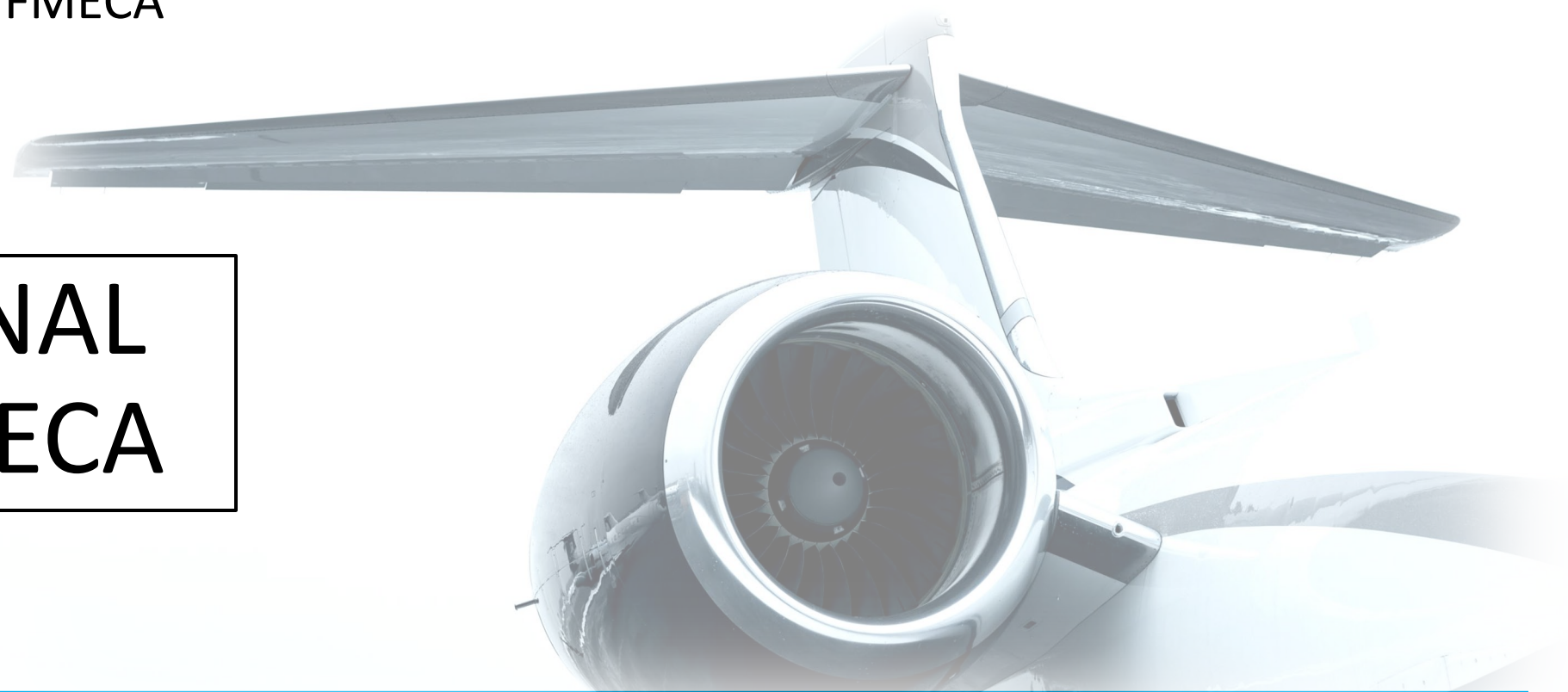


Safety and Reliability Training

Module 2: RELIABILITY ENGINEERING

Submodule 2.B: FMEA/FMECA

FUNCTIONAL FMEA/FMECA





Objectives of this training

- ✓ To learn the methodology of Functional FMEA





Outline

- ☐ FMEA Introduction
- ☐ FFMEA – Introduction
- ☐ FFMEA – Ground Rules and Assumptions
- ☐ FFMEA – Functional model
- ☐ FFMEA – Block Building
- ☐ FFMEA – Block Functions
- ☐ FFMEA – Block Reliability Model
- ☐ FFMEA – Block Failure Modes





Outline

- ❑ FFMEA – Block Failure Mode Effects
- ❑ FFMEA – Block Failure Mode Failure Rate
- ❑ FFMEA – Block Failure Mode Causes
- ❑ FFMEA – Block Failure Mode Detection
- ❑ FFMEA – Criticality Analysis
- ❑ FFMEA – Analysis Taxonomy
- ❑ FFMEA – Tools
- ❑ FFMEA Example



FFMEA – Introduction (I)

- What a Functional FMEA is?? Does it hurt?



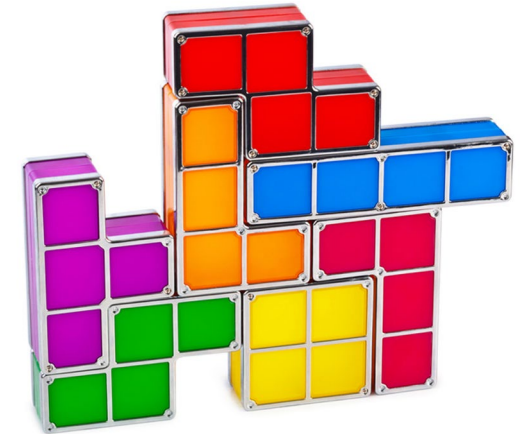
FFMEA – Introduction (II)

- The Failure Mode and Effect Analysis (FMEA) is a powerful dependability technique to evaluate systems and equipment inherent potentiality to fail leading to effects.
- FMEA methodology is based on the identification of the potential failure modes of each item according to analyst knowledge and engineering judgement, and their effects on the upper system levels.
- According to the analysis depth, FMEAs can be classified in:
 - Functional (FFMEA) – down to functional block level
 - Piece-Part (PPFMEA) – down to commercial hardware level



FFMEA – Introduction (III)

- A FFMEA analyses functional blocks whose scope and level of indenture is dependant on the analysis. So, neither predefined functions nor failure modes can be attributed as they can be in the case of PPFMEA.
- It is adequate:
 - When equipment or system design is not mature.
 - When the availability of time and/or resources for the analysis are limited.
 - As an initial analysis to be later completed by PPFMEA.



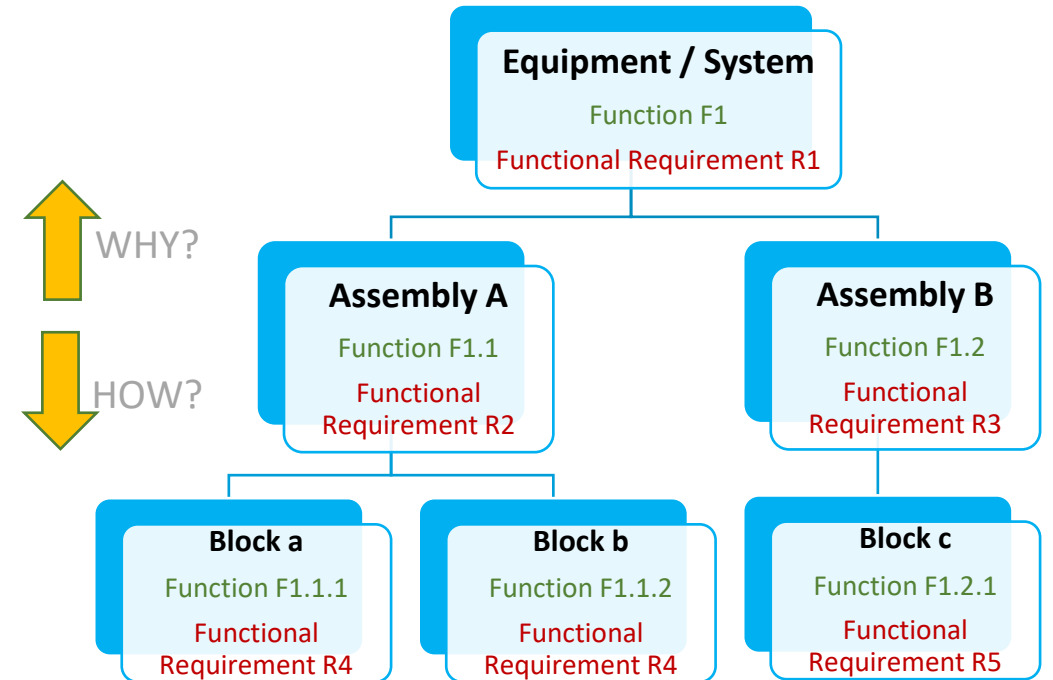
FFMEA – Grounds rules and Assumptions

- Similarly to other FMEAs, the analysis must be based on ground rules that determine its essential aspects, such as:
 - ✓ Team and responsibilities.
 - ✓ Purpose, according to client necessities.
 - ✓ Scope, including boundaries and interfaces.
 - ✓ Failure and success criteria.
 - ✓ Tool (spreadsheet template or WRR).
- Some assumptions related to the current project are normally required to tailor the analysis. Some examples are:
 - ✓ Taxonomy (it will be further explained later in this presentation).
 - ✓ Rating scales, if necessary (e.g. for severity, frequency...etc.).



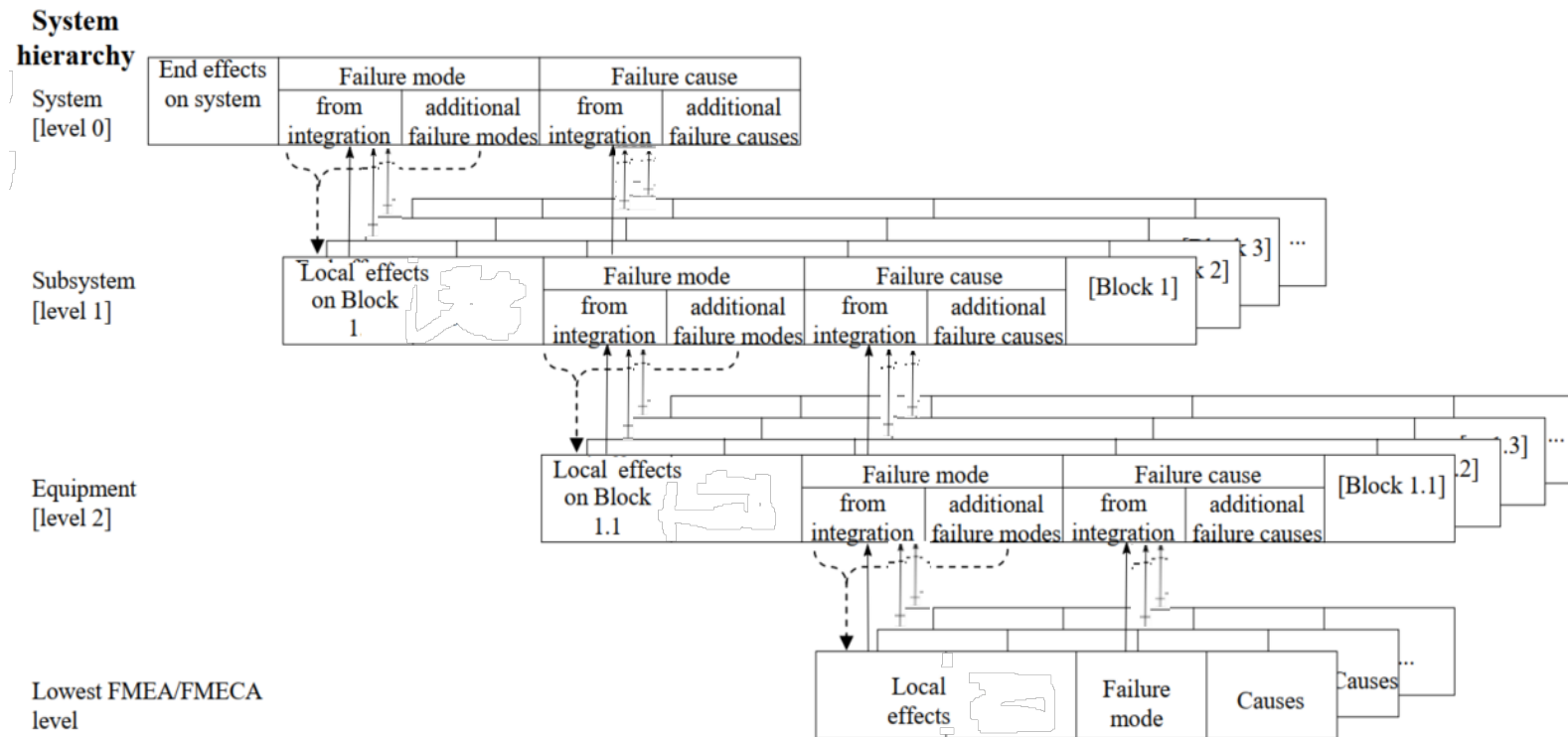
FFMEA – Functional model (1)

- The FFMEA model is based on a hierarchic breakdown of the equipment / system functions as conceived according to its corresponding requirements.
- Functions and requirements can be allocated to determine functional failures at lower levels.
- At each level, items can be physical ones (assemblies, modules) or functional (blocks).



FFMEA – Functional model (2)

- Similarly to other FMEAs, causes, failure modes and effects must keep their mutual relationships along the analysis.



FFMEA – Functional model (3)

- Several indenture levels are identified, each of them composed of hardware elements or functional blocks.
- Definition of levels and functional blocks should consider that:
 - ✓ A reduced quantity of them does not lead to an easier analysis since it would result in complex blocks, difficult to deal with.
 - ✓ On the other hand, a high number would make the FFMEA quite extensive and complex.
- Therefore, a trade-off between quantity and complexity must be reached for the sake of the analysis.



FFMEA – Block bluiding

- Blocks are the simplest FFMEA elements.
- Blocks can perform one or several functions and can include parts or not depending on design maturity.
- As a general rule for electronic equipment, a block is built from an integrated circuit, if any, and its related components. However, others such as those related to power decoupling or equipment interfaces are built because of their criticality.
- It is advisable that blocks in the same indenture level are independent, this is to say their failure probability are not affected among themselves, as an assumption at the least.



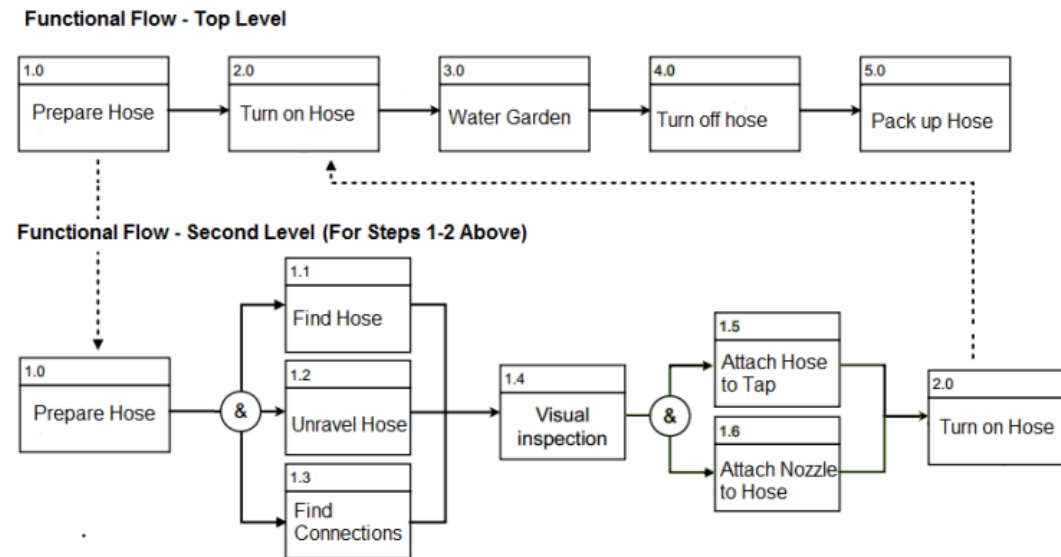
FFMEA – Block Functions

- Block functions can be established according to:
 - Allocated functions of the immediate upper-level block.
 - Input and output interfaces:
 - ✓ Electrical signals such as power supplies, grounds, analog / digital inputs or outputs...etc.,
 - ✓ Data connections like I2C, SPI...etc.
 - ✓ Physical interchange such as heating, cooling...etc.
 - ✓ Human interactions like visual, auditory...etc.
 - Presence of key components (microcircuits) inside which feature the block functionality.



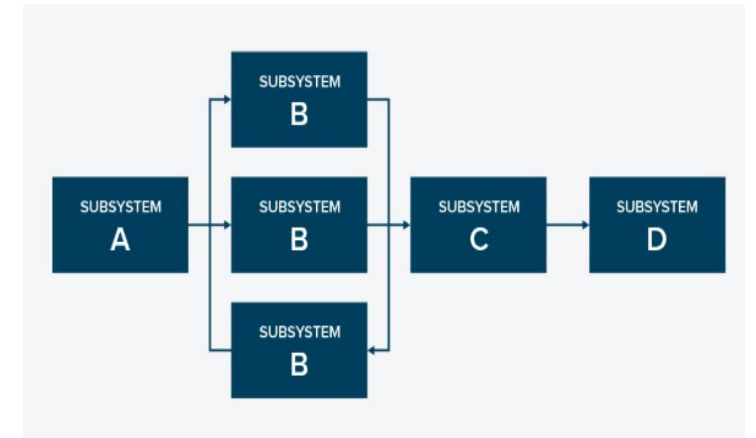
FFMEA – Block Building

- Block functions provide flexibility to the functional model since they act as an intermediate level between block and its constituent parts by establishing functional relationships among these ones.
- Block functions must be hierarchically coherent with those of the other blocks at upper, lower and same indenture level in the functional model.



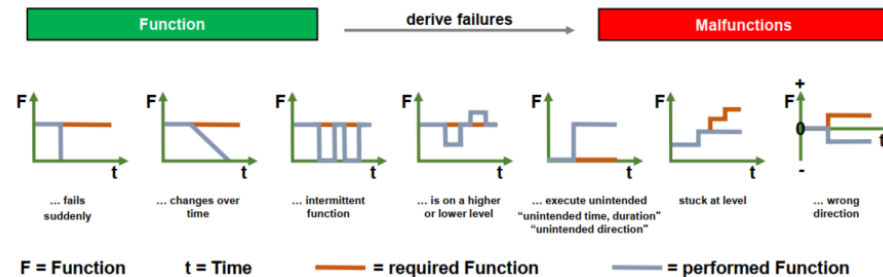
FFMEA – Block Reliability model

- It is very advisable to analyse the reliability relationships between blocks. This is to say, identify the functional path(s) across blocks which each high-level function requires to be accomplished.
- This task highlights:
 - Single point failures which may impede equipment functions
 - Functional redundancies (fail-tolerant architectures).
- Each functionally redundant channel should be modelled as a higher-level block composed by one or several lower-level ones.



FFMEA – Block Failure Modes

- Block failure modes are estimated according to common failures types:
 - ✓ Function is not available when requested.
 - ✓ Function stops while required.
 - ✓ Function is performed without being required.
 - ✓ Function performance is out of specification (late, degraded...etc.).
 - ✓ Function provides false or inaccurate data.
- Generally, requirements determine the functional failure threshold.



FFMEA – Block Failure Modes

- From the point of view of the analysis, Functional failures and Failure Modes concepts can be different since the first ones may result from external causes (e.g. output loss caused by missing power supply) whereas Failure Modes are due to internal ones (e.g. output loss caused by shorted power transformer).
- An adequate taxonomy of failure modes is highly advisable to keep analysis traceability and modularity.



FFMEA – Block Failure Mode Effects (1)

- Functional effects of each block failure mode are evaluated at several indenture levels:
 - LOCAL – related to the effects led in the output of the block (WRR© considers, specifically, the failure mode of the immediate upper-level one).
 - NEXT (optional) – related to the failure modes led in an intermediate upper block.
 - END - related to the failure modes of the highest level under analysis (equipment or system).



FFMEA – Block Failure Mode Effects (2)

- Special care must be taken to propagate coherently the failures effects upwards, rather than forwards, according to the functional model.
- However, in some case, (e.g. shorted input), there should be also taken into account the consequences on previous blocks when analysing. For example, these could avoid partially or totally powering up of other equipment area.
- An adequate taxonomy of failure effects is highly advisable to keep both analysis traceability and modularity.



FFMEA – Block Failure Mode Effects (3)

- Normally, each Failure Mode is linked to an effect. However, different effects can be assumed to occur in certain cases such as:
 - For avoiding an excessive quantity of block failure modes (e.g. blocks managing a great quantity of signals).
 - When effects depend on:
 - Equipment phases.
 - Equipment states (operating, non-operating,...etc.).
 - External events (transients, inrush currents, human behaviour,...etc.).
- In these cases, a conditional probability usually expressed as a percentage can be included in the effect FR computation:

$$\text{EffectFR} = \text{ModeFR} * P(E/M)$$



FFMEA – Block Failure Mode Failure Rate

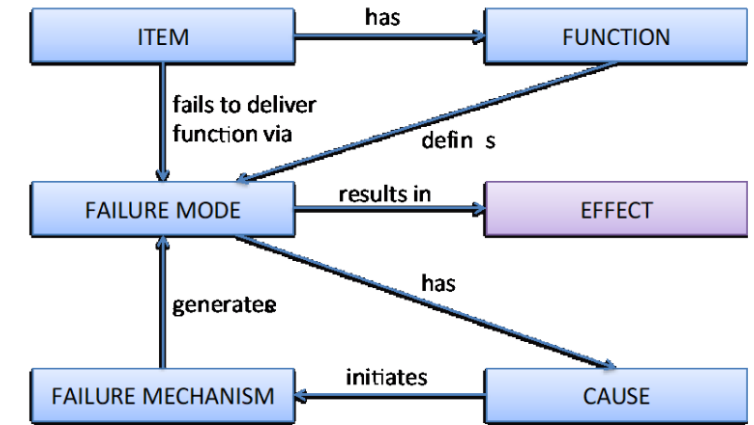
- Regarding to Block Failure Mode rate estimation, either without enough knowledge or by agreement, the following assumptions can be made:
 - ✓ Block Functions are proportionally weighted about the block.
 - ✓ Failure Modes are equally distributed for each block function.
 - ✓ Part failure rates are considered as independent.
 - ✓ Part failure modes are regarded as critical.
- As a result, Block Failure Mode Failure Rate can be computed like this:

$$\text{ModeFR} = \text{BlockFR} * \text{FunctionRatio} * \text{ModeRatio}$$



FFMEA – Block Failure Mode Causes

- Similarly to other FMEAs, immediate lower-level failure modes contribute to under-analysis failure mode occurrence as causes.
- Failure mode causes lead to failure mechanisms, such as “fatigue” or “corrosion” which in turn results in failure modes.
- Whether included or not in the PPFMEA, causes identification and mitigation allows for control of failure mode frequency.



FFMEA – Block Failure Mode Detection

- As far as possible, failure detection functions should be assigned to specific blocks in the functional model.
- These functions are not normally included in the main equipment ones. However, their functional failures can be relevant since:
 - ✓ Main Function Failures Modes can get undetected (latent).
 - ✓ Significant effects can be caused if related main function also fails. Multiple failures are, at first, out of the FFMEA scope. However, for the sake of following analyses, such as FTA, their potentiality should be clearly indicated.
 - ✓ Testability requirements (e.g. Failure Detection Rate) may be imposed.



FFMEA – Criticality Analysis

- Criticality analysis is a systematic approach to evaluate and prioritize assets based on the severity and likelihood of their potential failures modes.
- It is not common to include a Criticality analysis in the FFMEA we usually do.
- If required, any of the existing methodologies (RPN, Severity-frequency matrix,...etc.) can be used.

		Consequence Scale				
Prob-ability Index	Conse-quence Index	1	2	3	4	5
		1	2	3	4	5
Probability Scale	5	Crit, 13	Crit, 12, 11, 10, 9, 8, 7	Crit, 6, 5, 4	Crit, 3, 2	Crit, 1
	4	Crit, 14, 13	Crit, 12, 11, 10	Crit, 9, 8, 7	Crit, 6, 5, 4	Crit, 3, 2
	3	Crit, 15, 14, 13	Crit, 12	Crit, 12, 11, 10	Crit, 9, 8, 7	Crit, 6, 5, 4
	2	Crit, 17, 16	Crit, 15, 14, 13	Crit, 13	Crit, 12, 11, 10	Crit, 9, 8, 7
	1	Crit, 18	Crit, 17, 16	Crit, 15, 14	Crit, 13	Crit, 12, 11, 10



FFMEA – Analysis Taxonomy (1)

- An adequate taxonomy of the FFMEA objects; functions, blocks, failure modes, caused, effects and detection means is highly recommended to keep both analysis traceability and modularity.
- Some of the FFMEA taxonomy related techniques can be:
 - ✓ An efficient and scalable coding, considering both client necessities and ANZEN experience. For example:

B01
 B01.01
 B01.01F01
 B01.01F01M01
 B01.01F01M01E01
 ...etc.

WRR tool does not allow two items to have the coding so their scope should be included (system/equipment, assembly, block, part).



FFMEA – Analysis Taxonomy (2)

- ✓ Functions should be written as a simple noun phrase that defines what action has to be carried out.
- ✓ Failure modes should be expressed according to its function context (e.g. incorporating a word which denotes how it fails) without ambiguity. Requirements provide the function acceptable threshold.
- ✓ Failure effects (and causes, is applicable) are failure modes at higher (or lower) levels so same considerations apply.



FFMEA –Tools - Spreadsheet

- For non-complex FFMEAs, tailored spreadsheets can be prepared quickly from ANZEN existing templates which are based on previous experience.
- As a general rule, templates provide:
 - ✓ Fields for coding and denomination of analysis objects and other auxiliary for signals, comments and explanations.
 - ✓ Lists for avoiding different spelling
 - ✓ Calculations for quantitative results.



FFMEA –Tools - Spreadsheet

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O
	0 empty	0 empty 22 IN/D	0 empty 0 IN/D	0 empty 0 IN/D	0 empty 0 IN/D	0 empty	0 empty	0 empty 0 IN/D	8466 empty 0 IN/D	8466 empty 0 N/A R/Red Progress: 0.02%	8466 empty 0 IN/D	8466 empty 0 IN/D	8466 empty 0 IN/D	8466 empty 0 IN/D	8466 empty 0 IN/D
	Schematic Reference	Item/Component	Description	Failure Ref.	Component Type	PIN	Failure Mode	Failure Cause	Operational Mode	Failure Effect on Local Component	Failure Effect on Board	Failure Effect on System	Ref - Failure Effect on Local Component	Ref - Failure Effect on Board	Ref - Failure Effect on System
3	C1	Def: 470NF0402_6V3 0402 040260474MAT2A R10000 P/N: 011.809-PF	Function: Power Management Schematic Sheet: 10-FPGA_PWR	10.10.1.1	Ceramic Chip Capacitor	N/A	Drift	Ceramic Chip Capacitor DRIFT	Maintenance	No effect	No effect	No effect	AVIO2353-L3.10.1	AVIO2353-L2.10.1	FLAGS-L1.01.1
4	C1	Def: 470NF0402_6V3 0402 040260474MAT2A R10000 P/N: 011.809-PF	Function: Power Management Schematic Sheet: 10-FPGA_PWR	10.10.1.2	Ceramic Chip Capacitor	N/A	Shorted	Ceramic Chip Capacitor SHORTED							
5	C1	Def: 470NF0402_6V3 0402 040260474MAT2A R10000 P/N: 011.809-PF	Function: Power Management Schematic Sheet: 10-FPGA_PWR	10.10.1.3	Ceramic Chip Capacitor	N/A	Opened	Ceramic Chip Capacitor OPENED							
6	C10	Def: 470NF0402_6V3 0402 040260474MAT2A R10000 P/N: 011.809-PF	Function: ARINC, RSxxx Schematic Sheet: 13-FPGA_BANK_33_34_35	10.20.1.1	Ceramic Chip Capacitor	N/A	Drift	Ceramic Chip Capacitor DRIFT	Maintenance	No effect (Decoupling Capacitor)	No effect	No effect	AVIO2353-L3.10.4	AVIO2353-L2.10.1	FLAGS-L1.01.1
7	C10	Def: 470NF0402_6V3 0402 040260474MAT2A R10000 P/N: 011.809-PF	Function: ARINC, RSxxx Schematic Sheet: 13-FPGA_BANK_33_34_35	10.20.1.2	Ceramic Chip Capacitor	N/A	Shorted	Ceramic Chip Capacitor SHORTED							
8	C10	Def: 470NF0402_6V3 0402 040260474MAT2A R10000 P/N: 011.809-PF	Function: ARINC, RSxxx Schematic Sheet: 13-FPGA_BANK_33_34_35	10.20.1.3	Ceramic Chip Capacitor	N/A	Opened	Ceramic Chip Capacitor OPENED							
9	C100	Def: 100NF0402_50V 0402 C1005X7R1H104K R10000 P/N: 011.830-PF	Function: PCI Schematic Sheet: 22-PEX8718_IO	10.30.1.1	Ceramic Chip Capacitor	N/A	Drift	Ceramic Chip Capacitor DRIFT							
10	C100	Def: 100NF0402_50V 0402 C1005X7R1H104K R10000 P/N: 011.830-PF	Function: PCI Schematic Sheet: 22-PEX8718_IO	10.30.1.2	Ceramic Chip Capacitor	N/A	Shorted	Ceramic Chip Capacitor SHORTED							
11	C100	Def: 100NF0402_50V 0402 C1005X7R1H104K R10000 P/N: 011.830-PF	Function: PCI Schematic Sheet: 22-PEX8718_IO	10.30.1.3	Ceramic Chip Capacitor	N/A	Opened	Ceramic Chip Capacitor OPENED							
12	C101	Def: 100UF1206_6V3 1206 C1206C107M9PACTU R2000 P/N: 011.161-PF	Function: Power Management Schematic Sheet: 10-FPGA_PWR	10.10.2.1	Ceramic Chip Capacitor	N/A	Drift	Ceramic Chip Capacitor DRIFT							
13	C101	Def: 100UF1206_6V3 1206 C1206C107M9PACTU R2000 P/N: 011.161-PF	Function: Power Management Schematic Sheet: 10-FPGA_PWR	10.10.2.2	Ceramic Chip Capacitor	N/A	Shorted	Ceramic Chip Capacitor SHORTED							
14	C101	Def: 100UF1206_6V3 1206 C1206C107M9PACTU R2000 P/N: 011.161-PF	Function: Power Management Schematic Sheet: 10-FPGA_PWR	10.10.2.3	Ceramic Chip Capacitor	N/A	Opened	Ceramic Chip Capacitor OPENED							
15	C102	Def: 4.7UF0603_10V 0603 C0603C475MBPACTU R4000 P/N: 011.808-PF	Function: ARINC, RSxxx Schematic Sheet: 14-FPGA_BANK_12_13_14	10.20.2.1	Ceramic Chip Capacitor	N/A	Drift	Ceramic Chip Capacitor DRIFT							
		Def: 4.7UF0603_10V 0603 C0603C475MBPACTU	Function: ARINC, RSxxx					Ceramic Chip Capacitor							



FFMEA –Tools - WRR

- Windchill Risk & Reliability (WRR) is an integrated suite of software tools used for analyzing product performance and reliability. Some of FMEA module main features are:
 - ✓ FFMEA and PPFMEA capabilities.
 - ✓ FMEA building using either industry standards or user-defined ones.
 - ✓ Failure mode libraries for electronic and mechanical parts.
 - ✓ Automatic building (top-down and down-top) of FMEA levels and Fault Trees.
 - ✓ Error verification.
 - ✓ Built-in and user-defined calculations.
 - ✓ Reports and graphs generation according to user necessities.
 - ✓ Data links from Reliability Prediction.



FFMEA –Tools - WRR

PTC Windchill Quality Solutions - [My Training Project, System: Tablet PC]

File Edit View Insert Tools System FMEA Filter Window Help

Filter: No Filter --

FMEA Tree Items - Functional

Name	Part Number	System Tree Identifier	Reference Designator	Description	Manufacturer
Tablet PC		FMEA Tree1			
Processing Section		FMEA Tree2			
Touchpanel		FMEA Tree3			

System Tree Items FMEA Table FMEA Tree Items - Functional RBD Table RBD Blocks FTA Table Problems

FMEA Worksheet

Filter	Item / Process Function	Function / Process	Failure Mode	Mode Percentage	Local Effect	End Effect	Class	Cause of Failure	Current Controls	RPN	Recommended Actions	Responsibility	Target Date	Actions Taken
1			CPU failure	33,00	Motherboard halts processing	Tablet PC not functional		CPU fan failure	None	3	Create protection scheme to slow down CPU based on temperature.	Kevin - Engr.	17/07/2011	minor
2	Processing Section	Transfer data and provide computing functions	SRAM section failure	33,00	Microprocessor cannot store intermediate results	Tablet PC not functional		Metal to metal short	SRAM test sampling	3	Put additional controls on SRAM manufacturing line.	Tom - Mfg.	20/07/2011	Increased sampling of SRAM units produced
3			Video section failure	34,00	Video section provides degraded signal to touchpanel	Video appears with wavy lines		Video card loose	None	1	Add latch to lock video card to AGP	James - Engr.	25/07/2011	Added latch to AGP or motherboard
4			No response to touch	33,00	Touchpanel inoperable	Tablet PC not functional		Cable pin failure	Cable testing	0	Additional inspection of incoming goods	Pete - Receiving	24/01/2011	Added additional step to inspect cable pins
5	Touchpanel	Provide user access to Tablet PC applications	Inaccurate touch position	33,00	Loss of control	Limited operation		Linearity defect	None	0	Change cover application to eliminate need for connection to electrode pattern	Steve - Engrg.	07/01/2011	Redesigned glass substrate processing technique
6			Cursor stuck	34,00	Touchpanel inoperable	Tablet PC not functional		Short	None	0	None			None

Parts Table Prediction Data General Data FMEA Worksheet FMEA Control Plans FMEA DVPS FMEA Maintainability Data RBD Diagram FTA Diagram FRACAS Problems Form

Ready MIL-HDBK-217 FN2 FPMH

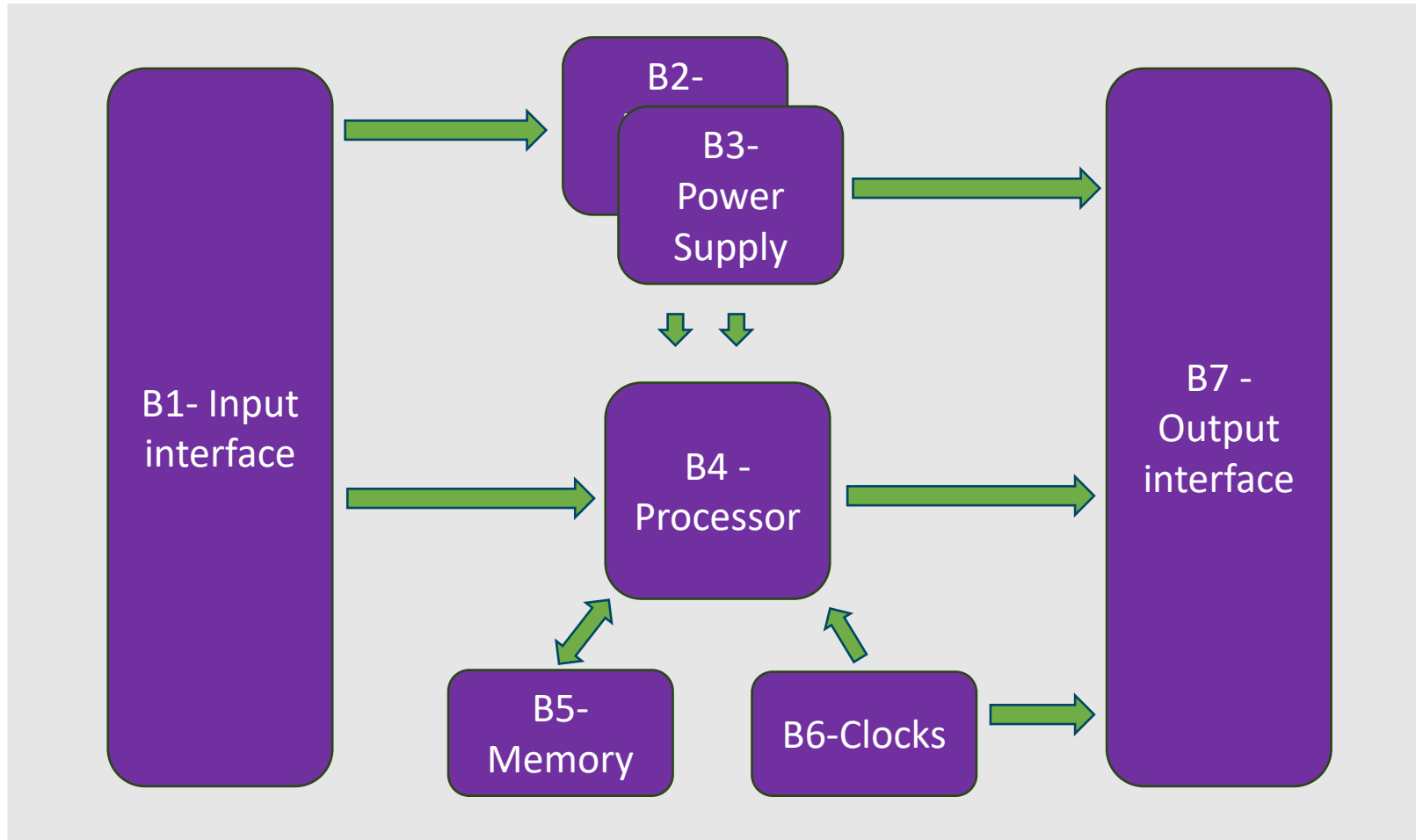


FFMEA Example – Introduction

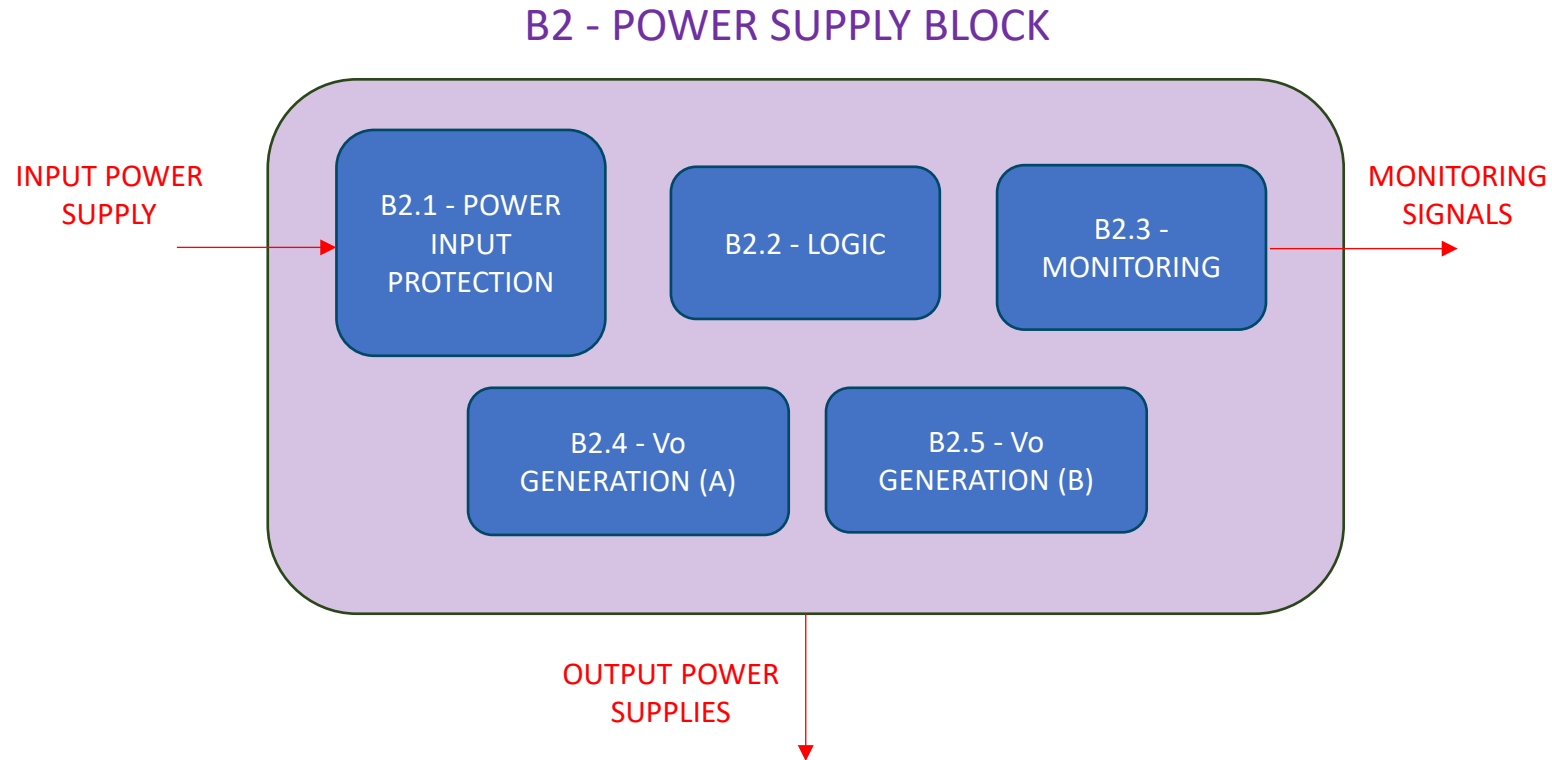
- The example FFMEA analyses a processing module preliminarily before being further completed by a PPFMEA.
- Design is mature to part level.
- Functional model is built from equipment knowledge, identifying four indenture levels:
 - 1º - Equipment
 - 2º - Subsystems
 - 3º - Functional Blocks
 - 4º - Parts
- Functions and signals are determined.



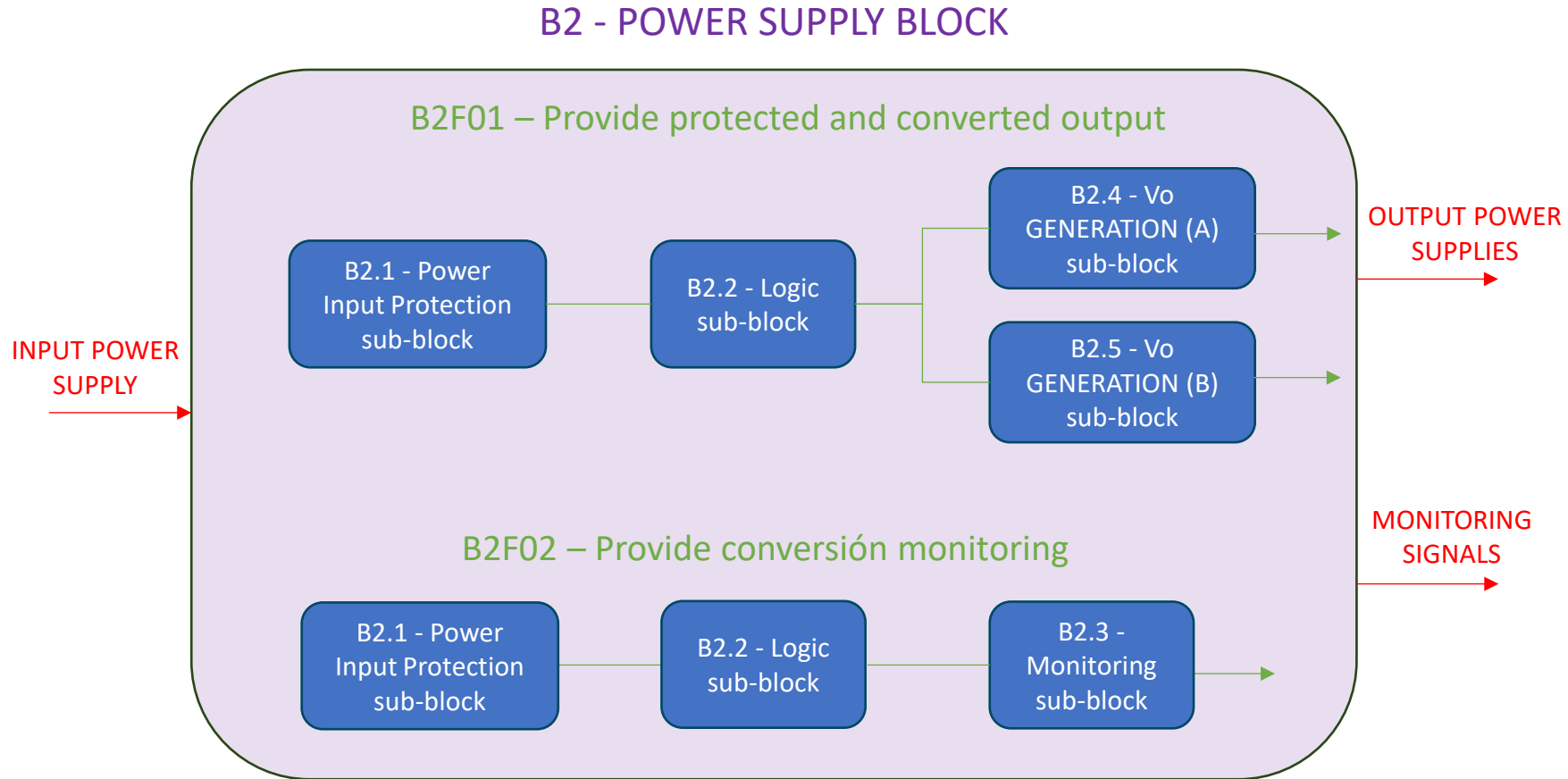
FFMEA Example – Indenture Level 1-2



FFMEA Example – Indenture Level 2-3

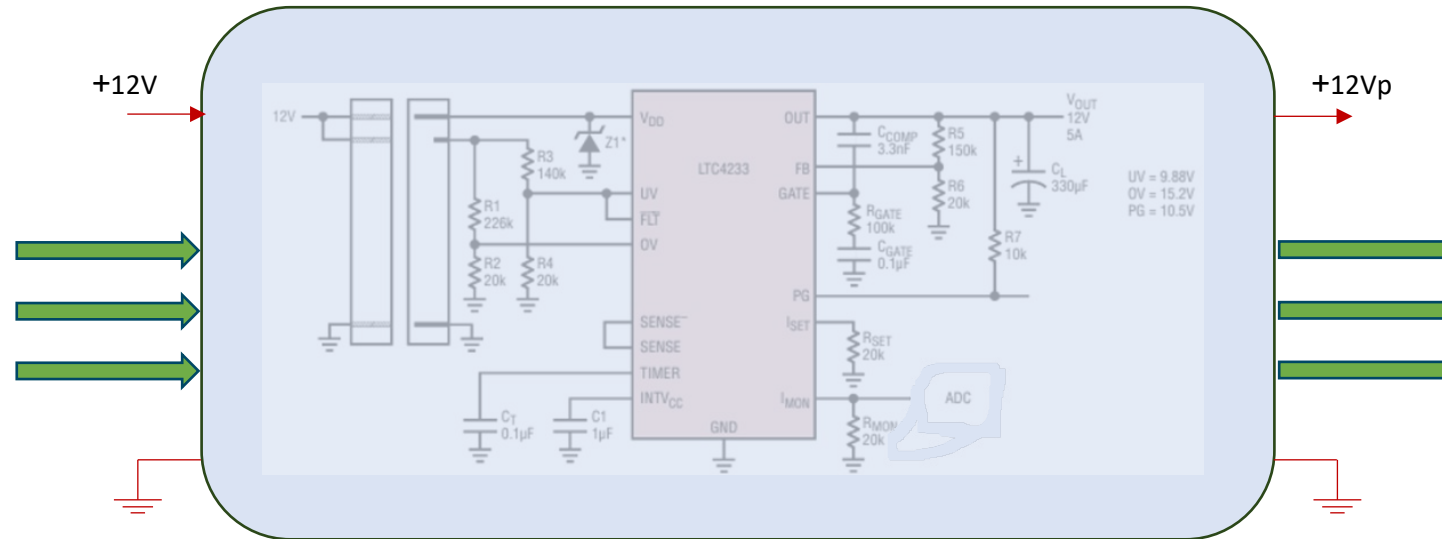


FFMEA Example – Reliability Model



FFMEA Example – Indenture Level 3-4(parts)

B2.1 - Power Input Protection sub-block



B2.1F01 – Protect against overvoltage

B2.1F02 – Protect against overcurrent

B2.1F03 – Provide thermal protection



FFMEA Example – Spreadsheet (1)

FFMEA Row Indicator	Block (L2) Code	Block (L2)	Function (L2) Code	Function (L2)	Sub-Block (L3) Code	Sub-Block (L3)	Sub-Block (L3) FR (fpmh)	Function (L3) Code	Function (L3)	Function (L3) Weight (%)
FFMEA001	B02	Power Supply	B2F01	Provide protected and converted output	B02.01	Power Input Protection	10,25	B2.1F01	Protect against overvoltage	100%
FFMEA002	B02	Power Supply	B2F01	Provide protected and converted output	B02.01	Power Input Protection	10,25	B2.1F02	Protect against overcurrent	100%
FFMEA003	B02	Power Supply	B2F01	Provide protected and converted output	B02.01	Power Input Protection	10,25	B2.1F03	Provide thermal protection	100%



FFMEA Example – Spreadsheet (2)

Failure Mode comment	Failure Mode Code	Failure Mode (L3)	Failure Mode (L3) Distribution (%)	Phases	Failure Mode Cause Code	Failure Mode Cause	Local Effect comment	Local Effect Code	Local Effect (L2)
when required	B2.1F01M01	No protection	50%	Taking off	B2.1F01M01C01	No U1 overvoltage detection	Circuit damage if overvoltage (worst case). No output. No monitoring.	B2F01M01	No output in B2 block
when required	B2.1F02M01	No protection	50%	Taking off	B2.1F01M02C01	No U1 overcurrent detection	Circuit damage if overcurrent (worst case). No output. No monitoring.	B2F01M01	No output in B2 block
when required	B2.1F03M01	No protection	50%	Taking off	B2.1F01M03C01	No U1 overheating detection	Circuit damage if overtemperature (worst case). No output. No monitoring.	B2F01M01	No output in B2 block



FFMEA Example – Spreadsheet (3)

End Effect comment	End Effect Code	End Effect (L1)	End Effect (L1) Conditional Probability (%)	End Effect (L1) rate (pmh)	Severity comment	Severity (1-10)
Alternative output of B3 block	F01M01	No effect	10%	0,51	No effect on operation	1
Alternative output of B3 block	F01M01	No effect	10%	0,51	No effect on operation	1
Alternative output of B3 block	F01M01	No effect	10%	0,51	No effect on operation	1

Detection means comment	Detection means	Detección Probability (%)	End Effect (L1) Criticality	Remarks
Detected by B4	Continuous built-in test	100%	001	No equipment function in case of B3.1 loss of output (double failure)
Detected by B4	Continuous built-in test	100%	001	No equipment function in case of B3.1 loss of output (double failure)
Detected by B4	Continuous built-in test	100%	001	No equipment function in case of B3.1 loss of output (double failure)



FFMEA Example – WRR (1)

Windchill Risk and Reliability - [FFMEA Training, System: Processing Module]

File Edit View Insert Tools System Filter Window Help

Project Navigator

Configurations

Filter	Configuration Identifier	Site, Configurat...	Quantity	Production Site	Shipment Date	Remarks
1	Reliability Prediction		#, #			
2	FFMEA		#, #			
3	All		#, #			
*	< Click here to insert a new record >					

System Tree Items Configurations FMEA Table FMEA Tree Items - System

FFMEA Worksheet



FFMEA Example – WRR (2)

Windchill Risk and Reliability - [FFMEA Training, System: Processing Module]

File Edit View Insert Tools System FMEA Filter Window Help

Project Navigator

FMEA Table

	FMEA Identifier	FMEA Type	Configuration	Remarks
1	FFMEA1	Component	FFMEA	FFMEA linked to Functional model (automatically linked failure rates)
2	FFMEA2	Functional		FFMEA linked to Functional model (not automatically linked failure rates)
*	< Click here to insert a new record >			

System Tree Items Configurations FMEA Table FMEA Tree Items - System

FFMEA Worksheet



FFMEA Example – WRR (3)

FMEA Tree Items - System

Filter

References

- ✓ AIAG VDA FMEA Handbook 5th ed.
- ✓ ECSS-Q-ST-30-02C 2009 – Failure modes, effects (and criticality) analysis (FMEA/FMECA)
- ✓ GSFC-HDBK-8004 – (NASA) Guideline for Failure Modes and Effects Analysis and Risk Assessment.
- ✓ IEC 60812:2018 Failure modes and effects analysis (FMEA and FMECA).
- ✓ MIL-STD-1629 - Procedures for Performing a Failure Modes, Effects, and Criticality Analysis.
- ✓ SAE ARP5580, Recommended Failure Modes and Effects Analysis (FMEA) Practices for Non-Automobile Applications



Thank you!!!

